Introduction to Electrical Engineering

Experiment 7 **Series and Parallel Resonance** **Spring 2019**

In this experiment, we will study series and parallel resonance in an LCR circuit. The circuit configurations for series and parallel resonant LCR circuits are given in **Fig. 7.1** and **Fig. 7.2** respectively. In both the configurations, the basic resonant circuit consists of a capacitance C, an inductance L and a resistance R. Most practical capacitors used in such circuits can be represented by a pure capacitance. Most practical inductors, on the other hand, are coils with a ferromagnetic core, which do not behave like a pure inductance, but have to be represented by a series combination of an inductance L and a resistance r_L. This resistance is not just the resistance of the wire constituting the coil; it also includes a resistance which represents the power losses in the coil due to hysteresis and eddy current. Hence the resistance r_L is always higher than the resistance of the coil as measured by a multimeter. Moreover, the nonlinear magnetisation characteristics of the ferrite core makes the coil behave in a slightly nonlinear fashion, leading to some distortions in the waveforms, as you may observe in this experiment.

Series Resonance gives a **maximum** current i_L through the inductor at a frequency called the **series resonant frequency**, denoted by f_s. As we can observe voltages only on a DSO, an external resistance r is added in the circuit, and the voltage $v_o = i_L r$ across the resistance r is considered as the output voltage. Thus the total resistance in the resonant circuit is given by $R = r_L + r$.

Parallel Resonance, on the other hand, gives a **maximum** voltage across the capacitor at a frequency called the **parallel resonant frequency**, denoted by f_p. The capacitor voltage is considered as the output voltage v_o in the parallel resonant circuit; so no external resistance is needed and $R = r_L$. An additional resistance R_S connected in series with the input voltage source to detect when the voltage v_o becomes **maximum** as frequency is varied.

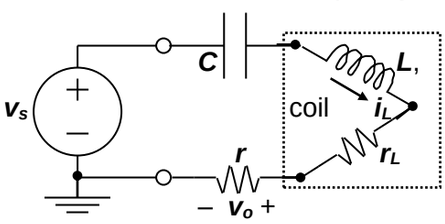

Fig. 7.1 Series Resonant LCR circuit

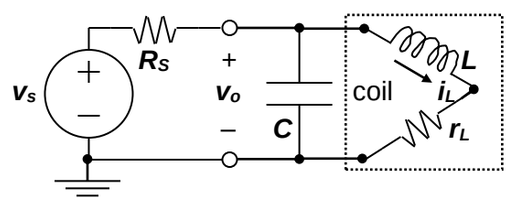

Fig. 7.2 Parallel Resonant LCR circuit

Series Resonance

In terms of the corresponding Phasors, the Transfer Function is given by

$$H(j\omega) = V_o / V_s = r / [(r + r_L) + j(\omega L - 1/\omega C)]$$

$$\Rightarrow |H(j\omega)|^2 = r^2 / [(r + r_L)^2 + (\omega L - 1/\omega C)^2].$$

$|H(j\omega)|$ will therefore be maximum when

$$\omega L - 1/\omega C = 0 \Rightarrow f_s = 1 / (2\pi\sqrt{LC}).$$

At this series resonant frequency,

$$H(j\omega) = H_0 = r / (r + r_L)$$

$\Rightarrow$ the phase angle of $H(j\omega) = 0^o$.

Parallel Resonance

In terms of the corresponding Phasors, the Transfer Function is given by

$$H(j\omega) = V_o / V_s = Z_p / (R_S + Z_p) = 1 / (1 + R_S / Z_p),$$

where Z_p is the impedance of the parallel combination of the coil and the capacitor:

$$Z_p = (j\omega L + r_L) / (1 + j\omega C r_L - \omega^2 LC) \cong j\omega L / (1 + j\omega C r_L - \omega^2 LC), \text{ if } \omega L >> r_L.$$

In most practical resonant circuits, this condition is satisfied, and hence

$$Z_p = L / C r_L \text{ (Real-valued) if } \omega^2 LC = 1.$$

This is also the maximum value of $|Z_p|$.

$|H(j\omega)|$ will therefore be maximum when

$$\omega L - 1/\omega C = 0 \Rightarrow f_p = 1 / (2\pi\sqrt{LC}).$$

As the measurements will be made at high frequencies near the radio-frequency range, one has to take into account the finite input impedance of the connecting wires and the DSO, which presents a 1MΩ resistance in parallel with a 11pF capacitance at each channel input. While 1MΩ is a very high resistance in comparison with other resistances in the circuit and can hence be treated as an open circuit, the capacitance of the connecting cable plus the input capacitance of the DSO can be quite significant in comparison with the actual capacitance used in the circuit. This capacitance, denoted by C_c, will therefore have to be determined and included in all calculations.

Experimental Work

1. Check the values of the given capacitor and resistors by noting their printed value and colour codes respectively. Set up the circuit shown in **Fig. 7.1**, with r = 10Ω. Apply the input voltage v_s from the WAVEGEN, and set it to have a sinusoidal waveform having frequency 100kHz and peak-to-peak value 2V.

2. Observe the waveforms of the voltages v_s and v_o on DSO Channels 1 and 2 and verify that they are both sinusoidal. This ensures that the circuit is indeed Linear and Time-invariant, in spite of the unavoidable non-linearity in the ferrite-core inductor.

3. Change the DSO display mode to the **x-y** mode, and note the elliptical display, indicating a phase difference between v_s and v_o. Increase the frequency gradually until the display becomes as close to a straight line as possible. At this point, v_s and v_o are in phase, implying that the voltages across the capacitance and the inductance cancel each other, resulting in the condition $H(j\omega) = r / (r + r_L)$. Note the value of this frequency and the slope of the straight line, giving the value of H_0.

4. Determine the value of L from the measured value of f_s, using the theoretical expression

   $$f_s = 1 / (2\pi\sqrt{LC})$$

   and the known value of C. Find the value of r_L from the measured value of $H(j\omega)$:

   $$H_0 = r / (r + r_L),$$

   using the known value of r. Measure the resistance of the coil with a multimeter and compare this value with the value of r_L just obtained by the measurement at resonance.

5. Change the mode of the DSO to **y-t** and determine the frequencies f_1 and f_2 at which the ratio of the AMPLITUDES of v_o and v_s becomes 70% of the value measured in step **3**. Note that near the resonant frequency, the impedance of the series resonant circuit is quite low, and its magnitude changes quite sharply as the frequency is varied. This results in a continuous variation in the amplitude of v_s due to the current drawn by the resonant circuit from the WAVEGEN ("**loading**"), though the WAVEGEN setting still continues to indicate the same value. You must therefore carefully measure the amplitude of both v_s and v_o at every step.

6. Find the value of the bandwidth $(f_2 - f_1)$ and hence the value of the Quality Factor Q, given by

   $$Q = f_s / (f_2 - f_1).$$

   Compare this value with the expected value

   $$Q = \sqrt{L / C} / R.$$

7. Insert R_s = 10kΩ to set up the Parallel Resonant circuit, and obtain a display of v_o against v_s with the DSO in the **x-y** mode. **Decrease** the frequency gradually until the display becomes a straight line again. Measure this slope, and hence find the value of $Z_p = R_p$ (Real-valued), using the theoretical expression

   $$H(j\omega) = 1 / (1 + R_S / Z_p),$$

8. Measure this frequency f_p, which is the resonant frequency of the inductance L and the parallel combination of C and the capacitance C_c:

   $$f_p = 1/ 2\pi \sqrt{LC_{eq}}, \text{ where } C_{eq} = C + C_c.$$

   It is easy to see that

   $$(f_s / f_p)^2 = 1 + C_c / C.$$

   Use this relationship to determine the value of C_c from the measured values of f_p, f_s and the known value of C. How does it compare with the DSO input capacitance?